

geobard • turn GeoJSON into prose

OPEN-SOURCE • FASTAPI • OPENAI-
COMPATIBLE

Turn GeoJSON into prose.

A small service that reads a map scene and writes what you'd see standing in it.

github.com/openfantasymap/geobard

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— A map is data. A place is a feeling.

A `FeatureCollection` is precise and machine-readable — and says nothing about **what it's like to be there.**

- ◆ Coordinates, types, names, geometry
- ◆ No "you're standing at the edge of a market square at first light"
- ◆ Players, agents, and readers want the **place**, not the table

geobard is the thin layer that closes that gap.

Open Fantasy Map · part of GaiaWM

— One small service. Four modes.

geobard is a thin, well-prompted layer between **your map data** and a **language model**.

Window view

Describe the scene as if you stand in it. Human language only.

POST /narrate/window

Ask the scene

Answer a question grounded in what's actually present.

POST /narrate/prompt

Image prompt

Rewrite the scene as a prompt for an image model.

POST /image/prompt

Three modes generate from data. A fourth — `/narrate/photo` — reads a real photo *through* it (coming up). Every call can override `model` and `temperature`.

— Mode 1 — the window view

In: a scene with a point of view, bearing, time of day, features.

```
{
  "geojson": {
    "pov": {"lat": 44.49, "lng": 11.34},
    "bearing": 90,
    "time_of_day": "07:30",
    "features": [
      {"properties": {
        "type": "buildings",
        "name": "market hall"}}
    ]
  },
  "detail_level": "medium"
}
```

Out: prose, no map-speak.

*You're looking east into the early light.
Just **across the way** the old market
hall squares off against the morning,
its long roofline catching the first of
the sun while the street in front is still
half in shade.*

DISTANCE IN HUMAN TERMS · VAGUE WHERE THE DATA IS

— Mode 2 — ask the scene

Pass a `prompt` alongside the scene. The model reasons over **what's present** — and stays honest about what isn't.

```
{  
  "geojson": { "...": "..."},  
  "prompt": "What's the quickest  
            way out on foot?"  
}
```

GROUNDED • CONVERSATIONAL • 1-2 PARAGRAPHS

From here your fastest line is to keep the market hall on your left and follow the lane it fronts onto — it bends toward the open ground you can just make out ahead, away from the tighter streets behind you.

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— Mode 3 — the image prompt

Same scene, retargeted for an **image-generation model** — architecture and city structure first, so the render is faithful to the *place*.

Ground-level view looking east at dawn down a cobbled market street. A long stone market hall dominates the right, its arcaded façade and pitched roof catching low golden light; narrow row houses recede on the left into soft morning shade. Medieval European town, wide-angle, photographic, soft volumetric light.

IMAGE_SYSTEM[] ADDS STYLE HINTS · DETAIL_LEVEL TUNES DENSITY

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— Mode 4 — read a photo *through* its data

A real **photo** carries appearance; the **data** carries meaning. Interpret one against the other — don't caption, and don't narrate the data alone.

```
{  
  "geojson": { "...same area..." },  
  "image_url": "https://.../tower.jpg",  
  "viewpoint": "ground",  
  "grounding": "loose"  
}
```

IDENTIFY → EXPLAIN → RECONCILE • NEEDS A VISION
MODEL • NEW IN 0.2

*The leaning tower on your left is the old **Harper watchpost**. The record remembers it as a lookout over the river road — but the boarded windows and the ivy tell a plainer story: **no one has watched here in years.***

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— How it works

```
GeoJSON scene  —▶  geobard  —▶  prose / image-prompt
  features      mode-specific  { "text": "...",
  pov, bearing  system + user   "model": "..."}
  time_of_day   prompt → chat
  custom props  completion
```

- ◆ **No magic** — careful per-mode prompts do the translation, the model writes
- ◆ **Non-standard GeoJSON fields** are read as extra context, not ignored
- ◆ Pure functions over `(geojson, options, client)` — framework-free, reusable

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— Bring your own model

geobard speaks the **OpenAI chat-completions API** — so it runs against whatever you already use.

Default	OpenRouter (<code>https://openrouter.ai/api/v1</code>)
Swap	set <code>GEOBARD_OPENAI_BASE_URL</code>
Works with	OpenAI · Ollama · Groq · Together · any compatible endpoint
Per request	override <code>model</code> and <code>temperature</code>

LOCAL WITH OLLAMA · HOSTED WITH OPENROUTER · PICK PER CALL

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— The API surface

ROUTE	DOES
POST /narrate/window	Window view. <code>geojson</code> , <code>system[]</code> , <code>detail_level</code> → <code>{text, model}</code>
POST /narrate/prompt	Answer a <code>prompt</code> about the scene
POST /narrate/photo	Interpret an <code>image_url</code> through same-area data (vision model)
POST /image/prompt	Image-gen prompt; <code>image_system[]</code> style hints
GET /healthz	<code>{"status":"ok"}</code> once env is valid (Docker healthcheck)

All bodies take optional `model` and `temperature`. Narration modes accept `system[]` prompt lines; window, image & photo modes take `detail_level`.

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— Running in two commands

1 • start the service

```
docker run --rm -p 8000:8000 \
  -e GEOBARD_OPENAI_API_KEY=sk-or-... \
  -e GEOBARD_OPENAI_MODEL=\
anthropic/claude-haiku-4.5 \
ghcr.io/openfantasymap/geobard:latest
```

2 • narrate a scene

```
curl -sS \
  localhost:8000/narrate/window \
  -H "content-type: application/json" \
  -d '{ "geojson": { ... },
        "detail_level": "medium" }'
# → { "text": "...", "model": "..." }
```

IMAGE: GHCR.IO/OPENFANTASYMAP/GEOBARD • MIT OR APACHE-2.0

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— Configuration

ENV	REQUIRED	DEFAULT
GEOBARD_OPENAI_API_KEY	yes	—
GEOBARD_OPENAI_MODEL	yes	—
GEOBARD_OPENAI_BASE_URL	no	openrouter.ai/api/v1
GEOBARD_TEMPERATURE	no	0.7
GEOBARD_HOST / GEOBARD_PORT	no	0.0.0.0 / 8000

Fail-fast: /healthz touches settings, so a missing key surfaces immediately.

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— Where geobard lives: GaiaWM

GaiaWM is world infrastructure for AI — a simulation-intelligence platform that layers space, time, society and causal influence over 40+ canonical worlds.



Spatial

40+ worlds, GIS precision — Toril, Star Wars, Middle-earth



Temporal

Time-aware state — history, seasons, cycles



Social

Factions, relationships, reputation



Influence

Causal simulation — channels & propagation



Narration →

geobard

Perception & description — geobard & EyeOnWorld

— geobard & EyeOnWorld — one engine

Click any location on the map — get an immersive description of standing there.

EyeOnWorld (live on **openfantasymaps.org**) is the same narration — geobard's window view — wired to real world data:

- ◆ Map tile → GeoJSON scene → `/narrate/window`
- ◆ The same `/image/prompt` mode renders the view
- ◆ Agents use `/narrate/prompt` to **reason over a place**

*Looking around, you're standing at the edge of a residential neighborhood where the midday sun beats down on the cobblestones. **Selduth Street** is just a few paces to your right...*

EyeOnWorld · generated from Waterdeep map data

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— Integrate it

- ◆ **As a service** — `docker run`, POST your scenes, get text back
- ◆ **As a library** — import the pure helpers from `geobard.llm` directly
- ◆ **In an agent loop** — `/narrate/prompt` gives spatial grounding over a scene
- ◆ **In a pipeline** — `/image/prompt` → your image model of choice
- ◆ **Self-hosted & private** — point the base URL at a local Ollama

Small surface, few dependencies — FastAPI, Pydantic, the OpenAI client.

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GEOBARD

A scene goes in. *Words* come out.

The open-source narration engine from GaiaWM — the engine behind EyeOnWorld.

github.com/openfantasymap/geobard · MIT or Apache-2.0

gaiawm.github.io · openfantasymaps.org · github.com/openfantasymap/geobard